



SC1006

Computer Organisation and Architecture

Computer System Overview

Prof Lin WeiSi
College of Computing and Data Science, NTU
Email: wslin@ntu.edu.sg

Sub-Systems in a Computer System & Presentation Schedule in Part 2

- Signal Chain Sub-System (**Week 8**)

- ADC-DAC
- Parallel and Serial (UART/SPI) Digital Interfaces
- Direct Memory Access Controller
- Interrupt Controller

- Memory Sub-System

- Semiconductor Memories (SRAM, DRAM, FLASH)
- Flash memory based Solid State Drives
- Magnetic Hard Disk Drives
- Cache Memory Management
- Virtual Memory Management

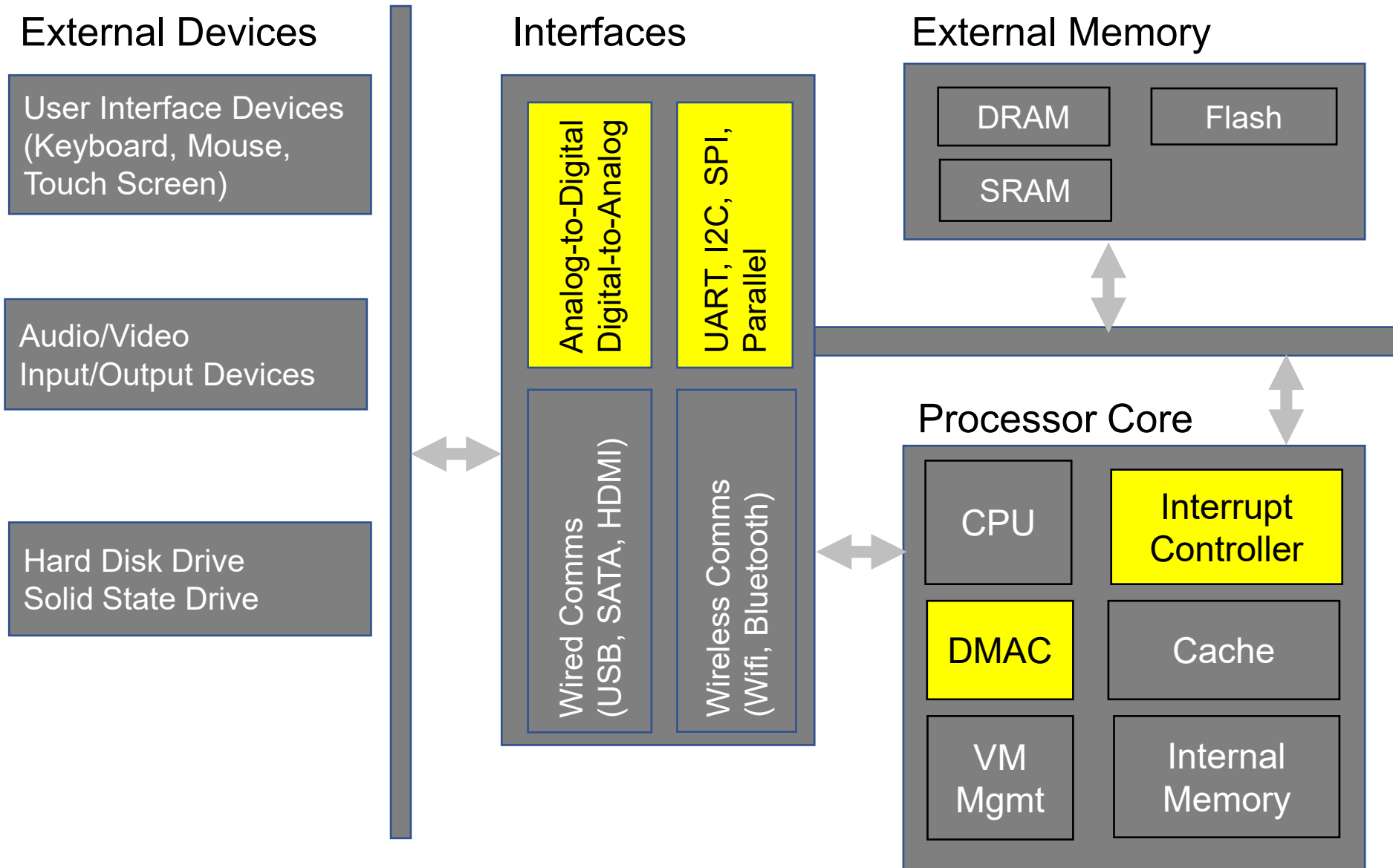
- Communication and User Interface Sub-System (brief intro only)

- Wired (USB, SATA, HDMI)
- Wireless (Wifi, Bluetooth)
- Input Devices (KBM, Capacitive Touch, Camera, Microphone)
- Output Devices (Display, Speakers)

- More for Processor Core

- Instruction Pipeline

Computer System Block Diagram



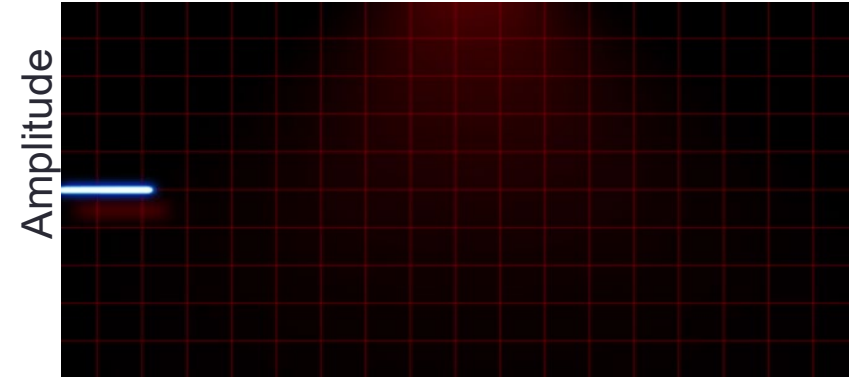
What is a Signal?

A **signal** is any **real-valued** function that changes in **time** and/or **space**.

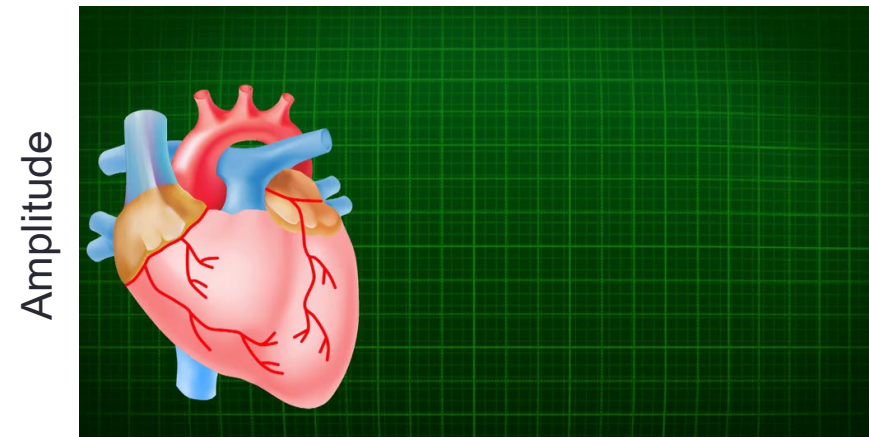
Signals that vary in time:



Waveform of audio/speech signal



Pulse signal



Electrocardiogram (ECG) signal

Cx1106

Computer Organization and Architecture

Interfacing to the Real World

Prof Lin Weisi

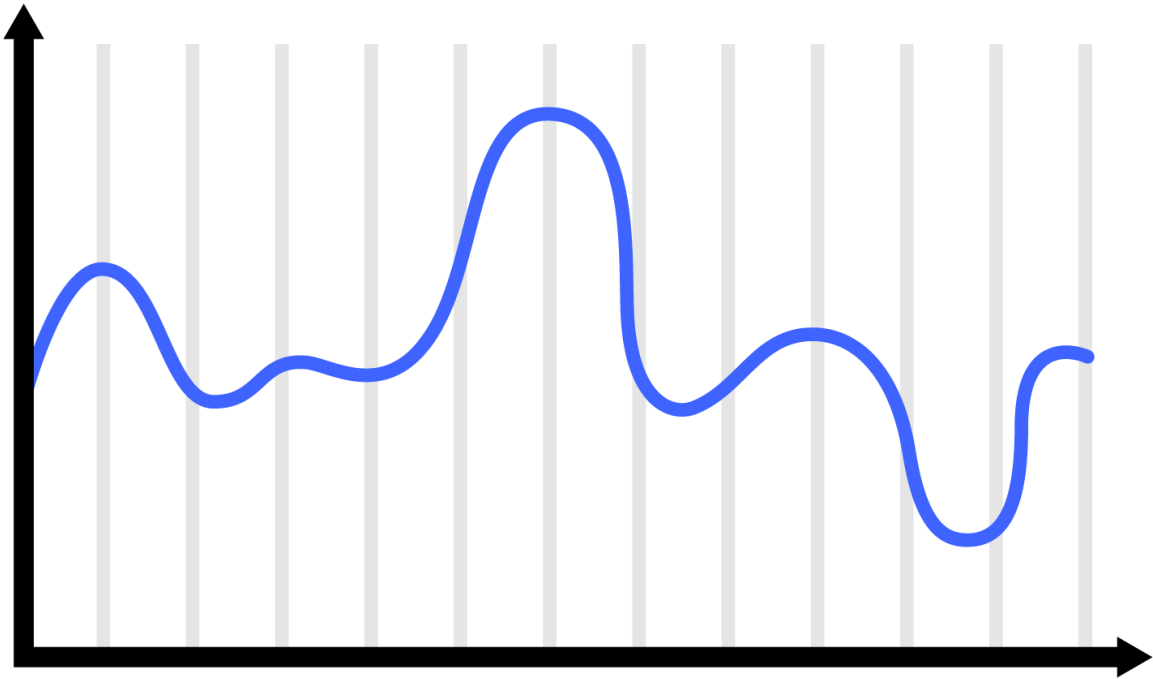
School of Computer Science and
Engineering, Nanyang Technological
University.

Email: wslin@ntu.edu.sg

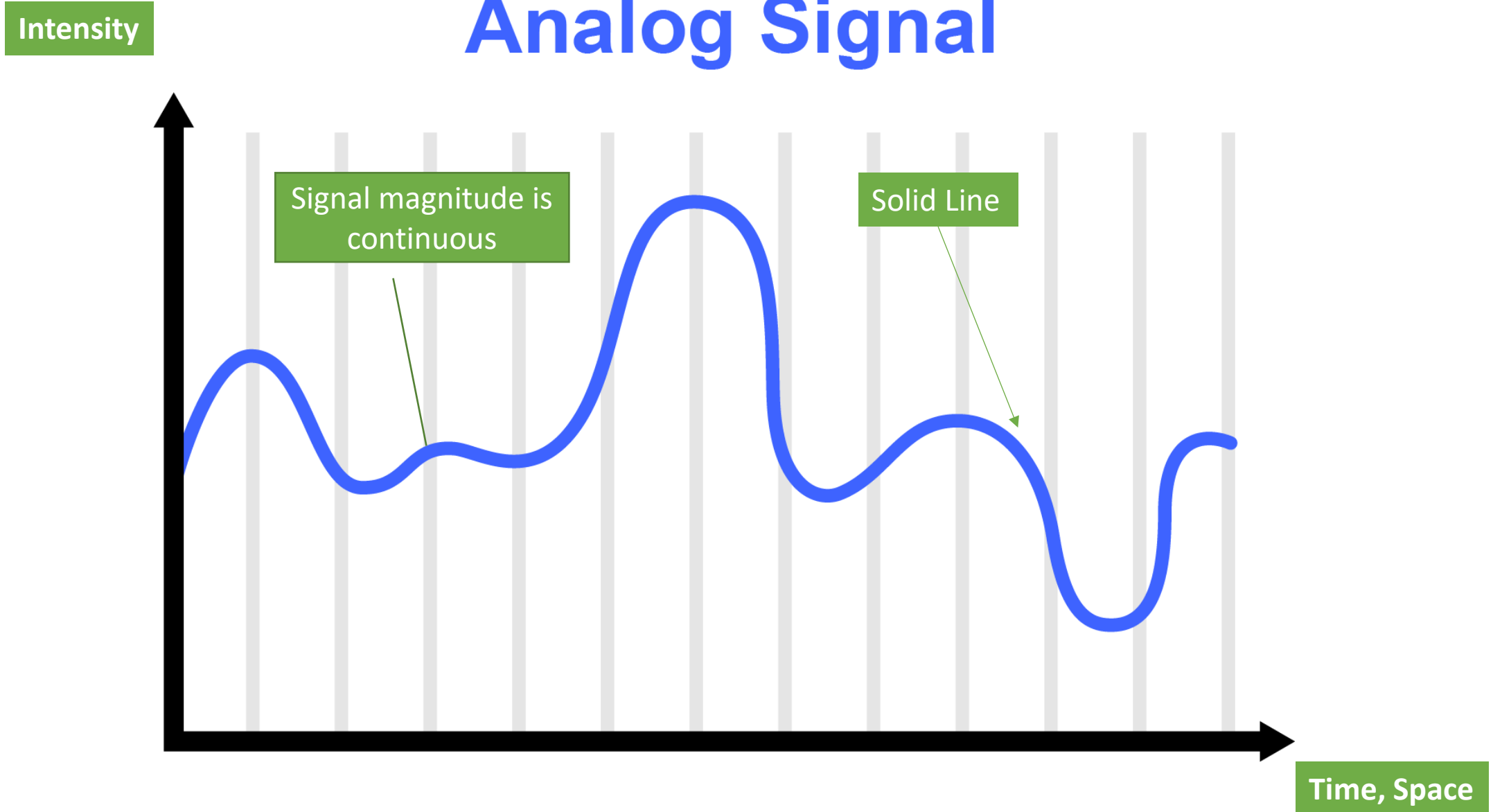


Analog and Digital Signal

Analog Signal

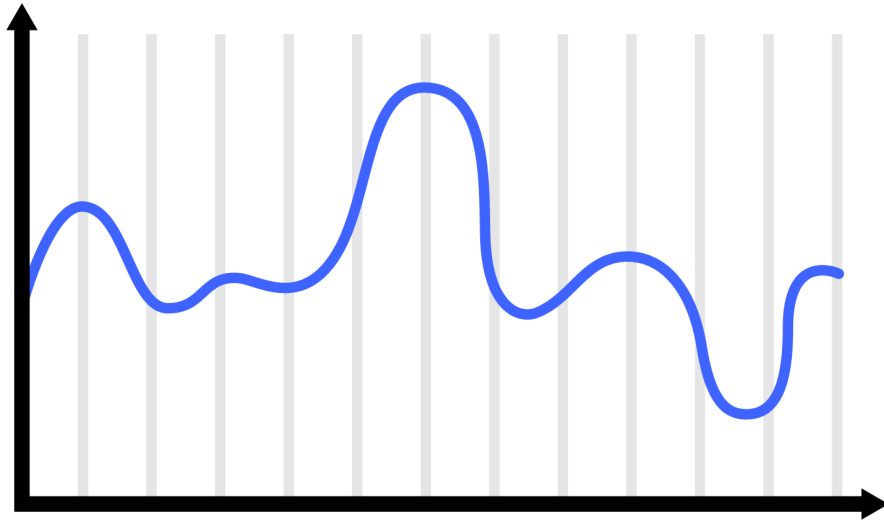


Analog Signal

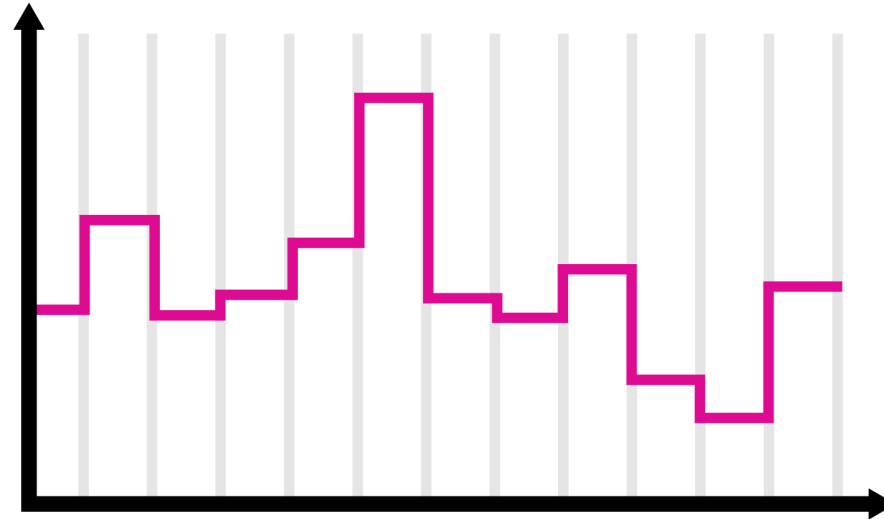


Analog and Digital Signal

Analog Signal

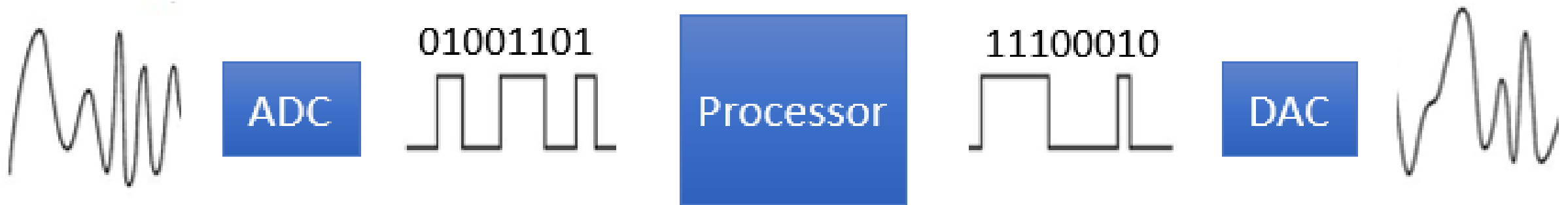


Digital Signal



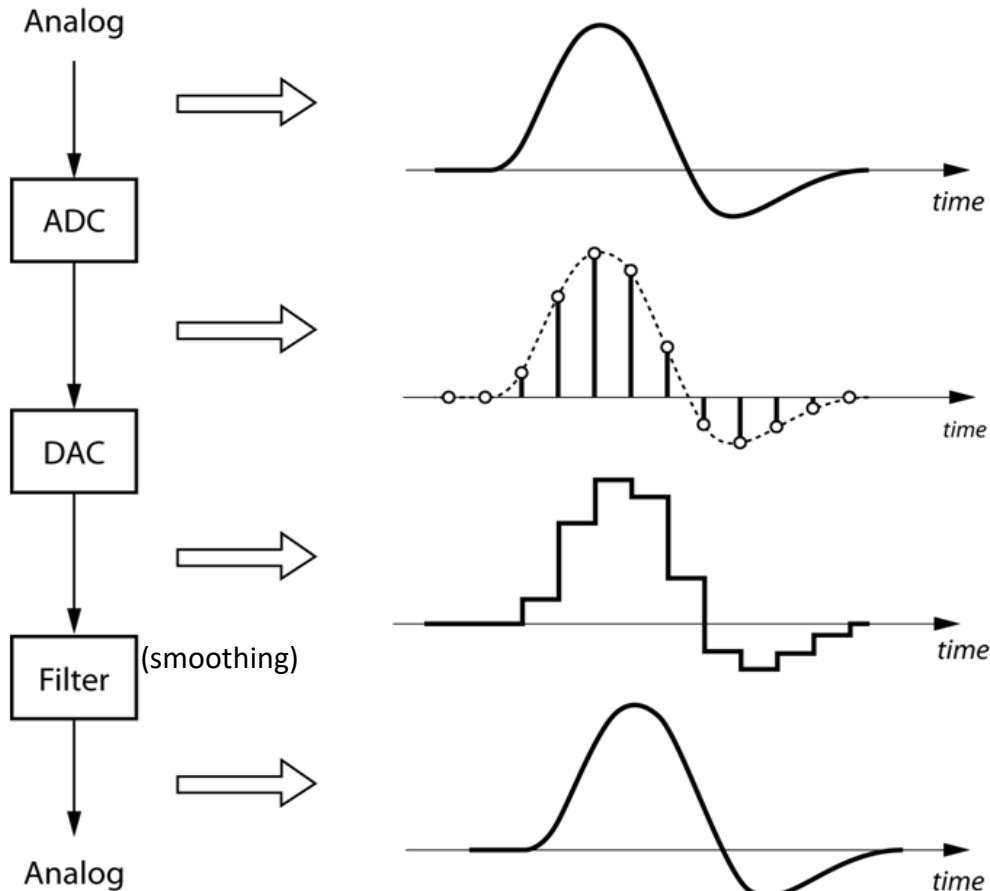
- Real world signals are **Analog** in nature.
 - Examples: **Sound, light, heat, pressure, smell, wind**, etc
 - In the past, most processing are done in analog domain.
 - With the introduction digital processors and decreasing cost to build them, digital processing became increasingly popular as it offer more **flexibility in implementation and are more tolerant to noise and component aging**.

Signal Chain Sub-System

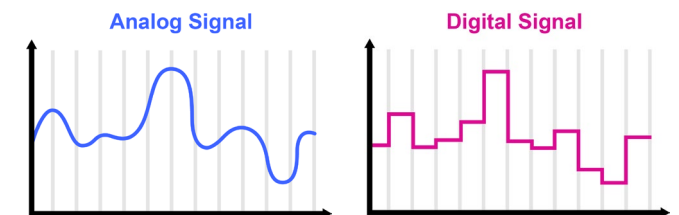


- Signal Chain Sub-System
 - ADC-DAC
 - Data transfer into and out of Processor

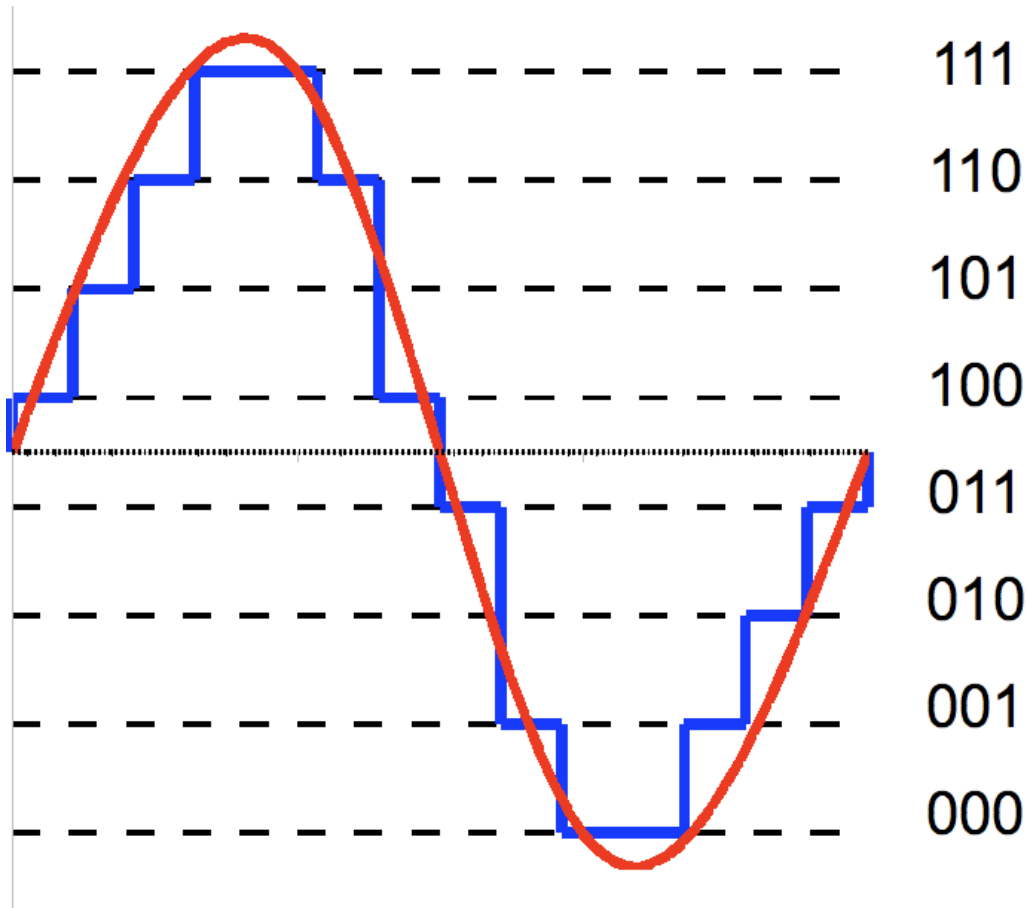
Transformation between Digital and Analog Domain



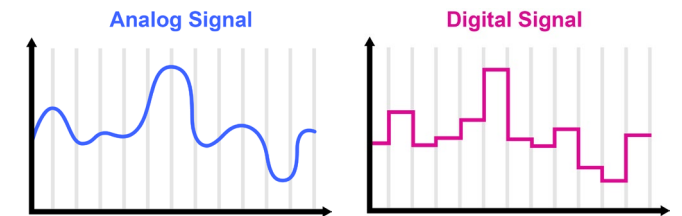
- Figure on the left shows **conversion between analog and digital domain**.
- Digital signal is obtained by **sampling the analog signal level at discrete time** (known as the **sampling interval**).
- Sampling frequency needs to be at least twice the signal frequency (**Nyquist theorem**).
- Typically, the analog signal level is assigned to the **nearest discrete voltage level allowed** in that particular digitization process.
- Analog signal can be reconstructed back by applying a smoothing filter after DAC.



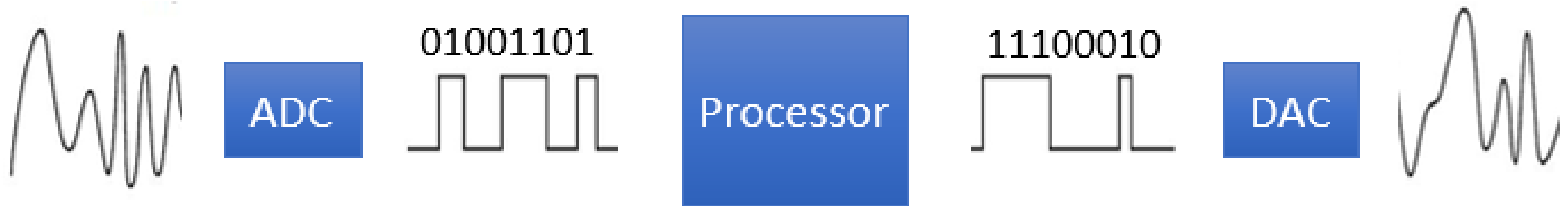
Digital Quantization



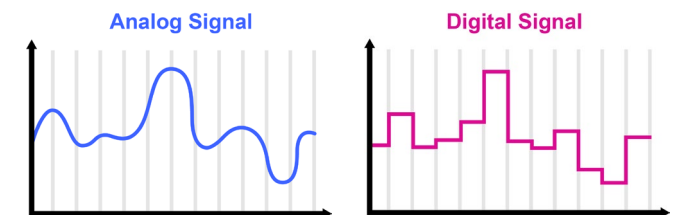
- Figure on the left shows a typical digital quantization process of an analog signal.
- A 3-bit system is used in this example. So all signal will be mapped to one of the 8 possible representations (000 to 111).
- At each sampling point, the analog signal level is approximated to the nearest digital equivalent.



Signal Chain between Processor and Real World



- Digital Processors **works only in the digital domain** so typical process these days is to convert the real-world analog signal to digital signal, allow the processor to work on the digital data, and reconstruct back the analog signal to be output to the real world.
- So **most of the interfaces we are dealing with in this course are in the digital domain.**



SC1006

Computer Organization and Architecture

Electrical Signal Interface

Prof Lin Weisi

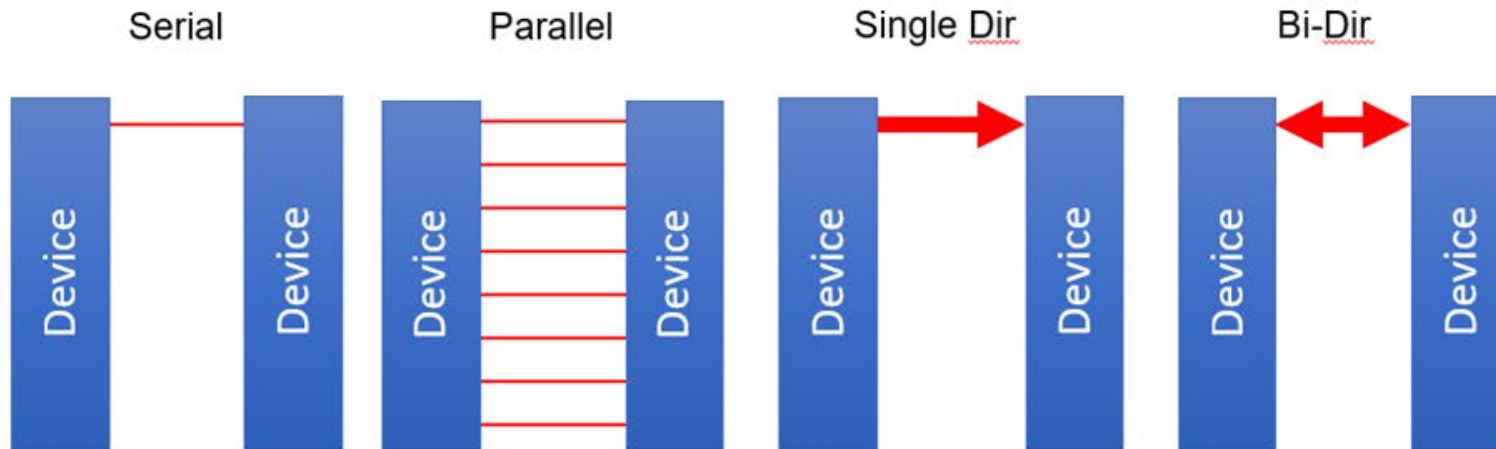
College of Computing and Data
Science, Nanyang Technological
University.

Email: wslin@ntu.edu.sg



Interface

- A **boundary** where two or more devices meet to **exchange information**. Some modes of connection below
 - Single-bit Data transfer (**Serial**)
 - Multiple-bits Data transfer (**Parallel**)
- For each mode above, the connection could be **Single direction** or **Bi-direction**.



Interface Compatibility

- Interfacing one electronic device to another requires compatibility in
 - **Electrical signal level** (to be explained in the slides next)
 - **Communication protocol** (Handshaking and Data signals; to be explained in the next two sub-sections).



Electrical Signal level (Safety)

- Primary consideration when connecting two electrical device together.
- Ensure that the output voltage level of output device do not exceed the maximum allowable input voltage level of the input device.
- Electronic devices will either get 'fried' i.e. spoilt or have its reliability reduced if the input voltage is higher than what they are designed for.
- So do check the voltage level which the device input/output is operating on (1.8V, 3V, 5V, 15V etc) before connecting them together.



Electrical Signal Level (Digital Data Transfer)

- A voltage as input: Logic '1' or '0' ?
- Four parameters
 - **VOH.**
 - Transmitting a Logic '1' yield a signal level of $\geq$ VOH
 - **VOL.**
 - Transmitting a Logic '0' yield a signal level of $\leq$ VOL
 - **VIH.**
 - A received signal with level $\geq$ VIH will be recognized as a Logic '1'
 - **VIL.**
 - A received signal with level $\leq$ VIL will be recognized as a Logic '0'
- These are for digital logic level compatibility.



Electrical Signal Level (Digital Data Transfer)



- Device #1
 - Device #1 datasheets: $V_{OH} = 2.4V$, $V_{OL} = 0.4V$
 - Output Voltage level for logic '1' = $V1+ \geq V_{OH} = 2.4V$
 - Output Voltage level for logic '0' = $V1- \leq V_{OL} = 0.4V$
- Device #2
 - Device #2 datasheets: $V_{IH} = 2.0V$, $V_{IL} = 0.8V$

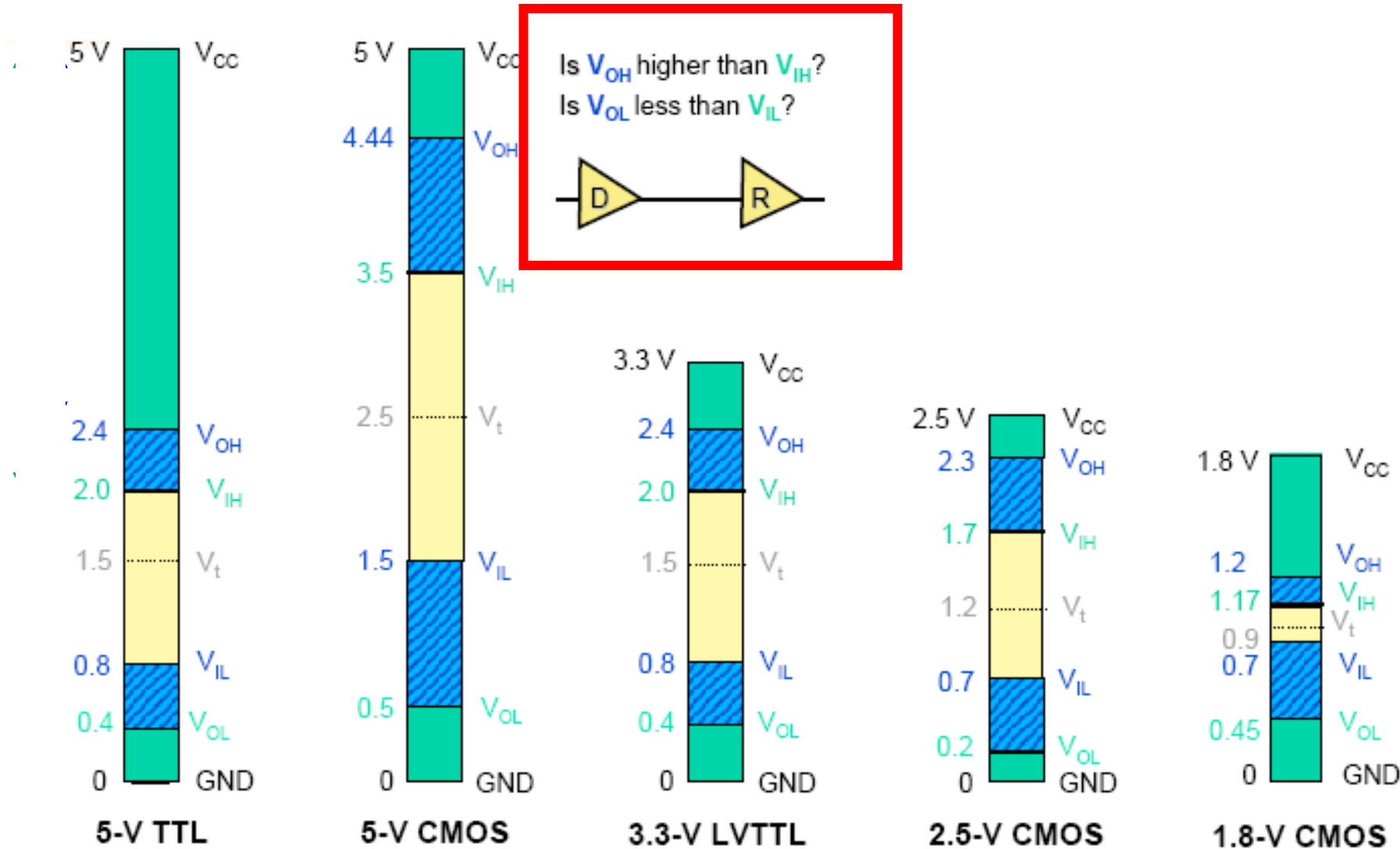
Device #2 senses the logic level correctly, because

$$V1+ \geq 2.4V > V_{IH} = 2.0V$$

$$V1- \leq 0.4V < V_{IL} = 0.8V$$

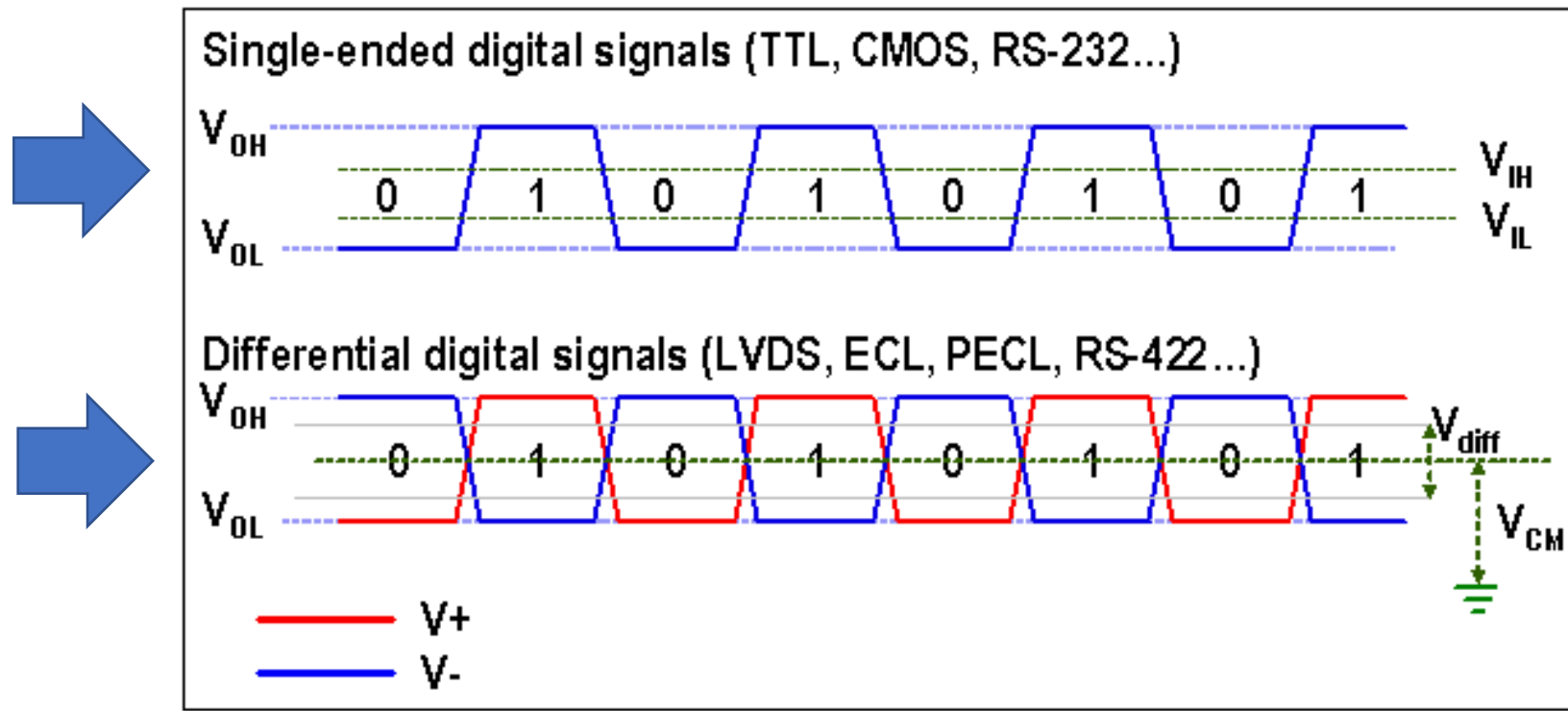


Electrical Signal Level Standards



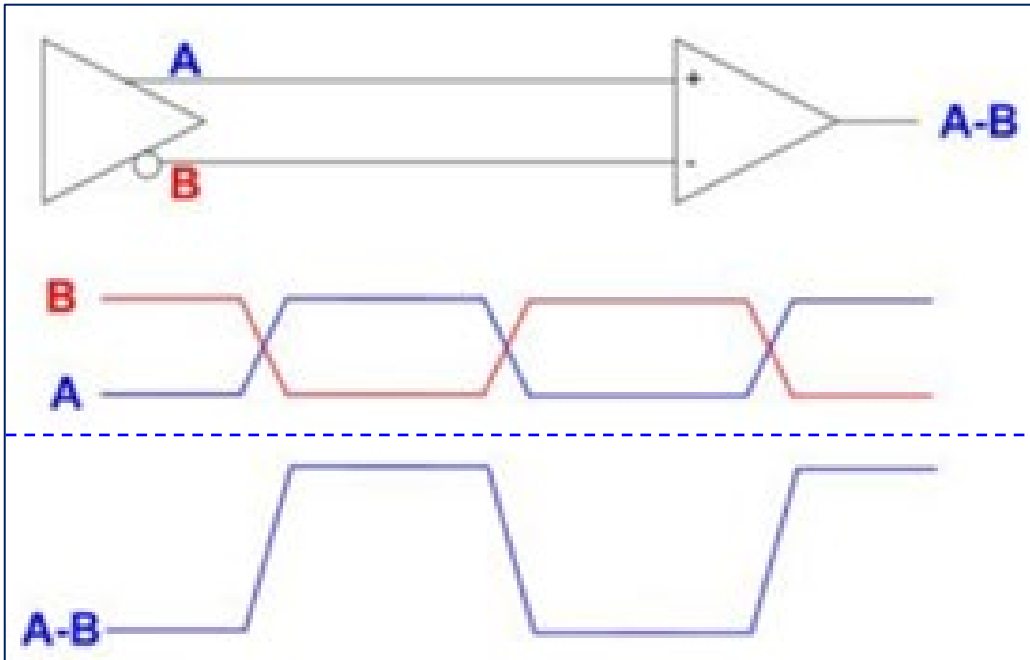
Differential Signals

- Differential signals has better noise tolerance so is able to be clocked at higher frequency.
- V_{CM} = Common Mode Ground.



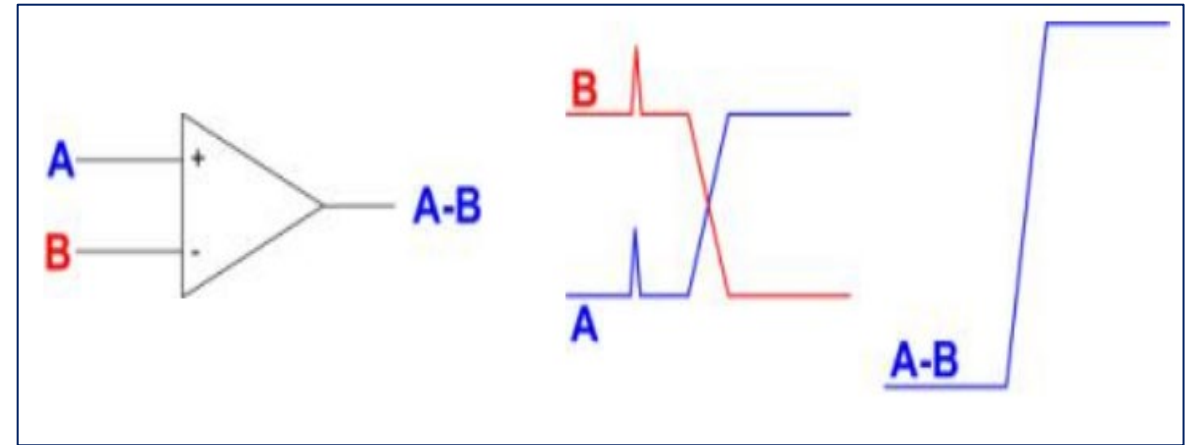
Differential Signals

Differential signals have better noise tolerance



A Larger Margin

A denotes V+
B denotes V-



External interference canceled